

WELL COMPLETION REPORT (SYNTHETIC)

OIL-DGB-011 - Digboi-011

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000121
Field / District	Digboi / Tinsukia
Block	PEL-ASSAM-01
Basin	Assam-Arakan
Formation	Tipam (TIP)
Spud / Completion	2005-06-14 / 2005-10-28
Measured / TVD	2873 m / 2838 m
Trajectory	Directional; inc 18.0 deg; az 160.1 deg
Formation Interval	1378-2445 m
Lithology	Fine-medium sandstone with shale interbeds
Pore Pressure Reference	2803 psi
Fracture Gradient	15.44 ppg
Mud System	PHPA Water-Based Mud
NWIS Well Risk	Moderate (49.2/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	718	Surface
9 5/8 in casing	1666	Intermediate
7 in liner/casing	2528	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2005-08-16 1464 m	Mud Loss	Critical	Flow-out dropped below flow-in with pit loss. Mitigation: Optimize hydraulics and maintain conservative ROP through loss interval. Lesson: Use conservative hydraulics when re-entering equivalent loss-prone interval.
2005-08-24 1625 m	Kick	Critical	Unexpected background gas rise preceded influx. Mitigation: Re-establish overbalance and monitor pit/flow balance. Lesson: Maintain mud-weight window using offset pressure evidence.
2005-08-26 1659 m	Fishing Operation	Critical	Normal pipe-recovery attempts were unsuccessful. Mitigation: Maintain clear event chronology and NPT accounting during recovery. Lesson: Maintain detailed BHA inventory for rapid recovery planning.
2005-07-13 750 m	Mud Loss	High	Partial returns observed at shale shaker. Mitigation: Spot graded LCM pill and circulate at reduced rate. Lesson: Prepare LCM package before approaching correlated offset-well loss depth.
2005-09-27 2279 m	NPT	High	No depth progress occurred during the recorded interval. Mitigation: Use preventive maintenance triggers from repeated downtime patterns. Lesson: Consistent NPT coding enables meaningful cross-well comparison.
2005-10-22 2753 m	Pressure Spike	High	Circulating pressure climbed without equivalent torque change. Mitigation: Use pressure trend thresholds referenced to depth and flow rate. Lesson: Rate-of-change in pressure is more informative than a fixed threshold.
2005-07-03 550 m	Pressure Spike	Medium	Short-duration high-pressure excursion was recorded. Mitigation: Condition hole and confirm pressure normalization. Lesson: Rate-of-change in pressure is more informative than a fixed threshold.

Completion Summary

The synthetic well reached 2873 m MD in the Tipam interval. The NWIS knowledge record links 16 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.